

What is a URL

- Summary
 - URLs: the address of a unique resource on the Internet
 - Web servers handle URLs + resources
- Anatomy of a URL
 - Parts: scheme, domain name, port, authority, path to resource, parameters, and anchor
- Scheme
 - Indicates protocol
 - Protocol: method for exchanging + transferring data
- Authority
 - Domain: requested web server
 - Port: technical gate to resources on web server
- Path to Resource
 - path to resource on web server
- Parameters
 - Key-value pair list separated by &
 - Do extra stuff before returning resource
- Anchor
 - Provides directions to show content at indicated spot
- How to Use URLs
 - HTML can insert links, video, images, sound, etc.
- Absolute URLs vs. Relative URLs
 - Differentiate by looking at path of the URL
 - Absolute has full path and Relative has truncated path
- URL Usernames and Passwords
 - Used to immediately sign into websites using https security
- Semantic URLs
 - Use easy to understand words so anyone can understand them
- Data URLs
 - Allow small files to be embedded inline

What is a web server?

- Summary
 - Static web server: server send files as is
 - Dynamic web server: server sends updated, hosted files
- Deeper Dive
 - Hosting Files
 - Store all files on a dedicated web server
 - Communicating through HTTP

- Textual: plain-text commands for human readability
- Stateless: server and client don't remember previous communication
- Static vs Dynamic
 - Server can return both static and dynamic content

1. Stateless web servers are more popular in modern web apps because most apps require a login. By not being able to remember past communications, it prevents unauthorized users from accessing your account.
2. An http request I can make to the CSUMB server while using Canvas is viewing my inbox.
 - The path is /conversations
 - No parameters
3. One possible response from a server is loading the requested web content. Another response could be an error message displaying that the content could not be found. 404 would be the status for the latter situation and 200 would be the status for the former.